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Abstract

Cryptography is an important tool for keeping information secure in today's computer networks. As sensitive data is exposed to ever increasing cyber risks and unauthorized access, the conventional ways of safeguarding this data are no longer good enough. The purpose of this assignment is to gain an understanding of cryptographic techniques and to create a better encryption scheme based on what we learn in class, that is better than classical encryption.

This is the two-part course in the cryptology, where aspects of cryptography system integrate symmetric and asymmetric encryption methodology, and the design of a novel cryptographic algorithm based on the classical Caesar Cipher. The classical Caesar Cipher has been confronted with several security drawbacks, which include fixed shifting and brute-force attacks. These security drawbacks are addressed by developing a novel cryptographic algorithm based on the Caesar Cipher concept of dynamic shifting and the use of ASCII operations and logic.

The developed algorithm was tested by various test cases to ensure the correctness and efficacy of the algorithm. From the results, the modified algorithm exhibits improved security features compared to the conventional Caesar Cipher algorithm. This coursework emphasizes the need to focus on the development of cryptography to enhance information security systems.

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1. Introduction

In today's world information security has become very essential because of the growth and dependency in digital systems. Organizations and individuals mostly store and transmit digitally which makes them vulnerable to various cyber-attack. Attackers can steal, modify, or destroy the digital data if it is not protected properly. This has led to more cyber-crimes such as data breach, hacking, and unauthorized access increasing rapidly every day across the world. So, protecting sensitive personal information, financial information, and organizational data is a need in this modern computing systems.

The CIA triad the fundamental model of information security which stands for Confidentiality, Integrity and Availability. Confidentiality ensures that the data is only accessed only by the authorized users. Integrity ensures the accuracy of that data and that it remains unchanged. Availability ensures that the system and the data are accessible by the users when needed. Cryptography mainly supports data confidentiality and data integrity (Martin, 2025).

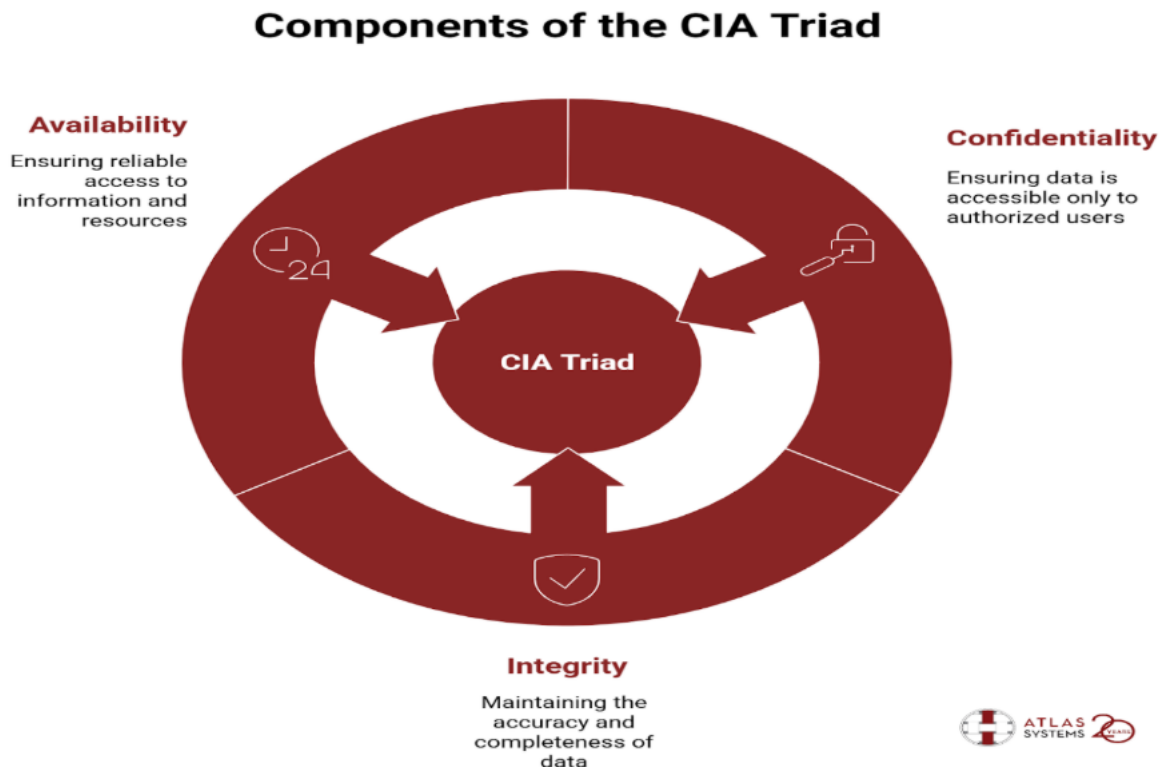


Figure 1: CIA Triad

Cryptography is the process of protecting data which uses mathematical concepts. It hides the original meaning of an information—converts a readable information (plaintext) into an encrypted unreadable information (GeeksforGeeks, 2025).

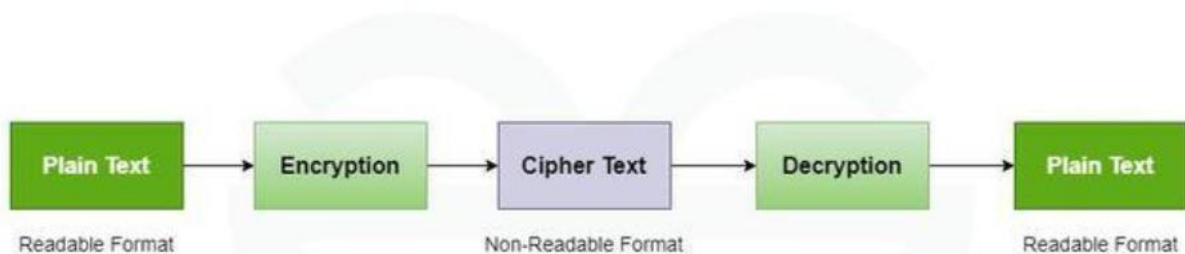


Figure 2: Cryptography

1.1 Cryptographic Terms

Encryption – It is a process transforming the readable plaintext into a secret scrambled text which someone with a key can only read.

The readable and unencrypted text is known as plaintext and encrypted text is known as ciphertext.

Decryption – It is like the opposite of encryption, the process of turning the secret scrambled text back to something readable.

Key – When you encrypt a data a key is what to need to decrypt it into a readable message. If you don't have a key, you cannot decipher and access the data. There are two main types of keys: Symmetric key and Asymmetric key (pCloud, 2025).

1.2 Symmetric and Asymmetric Encryption

Symmetric Key Encryption – Symmetric Key Encryption or also known as Single-Key encryption basically means using the same key for both encrypting and decrypting the sensitive data. This encryption method's first step is to create a key that all the involved individuals must keep a secret. This procedure uses both mathematical and logical operators to change the plaintext into a ciphertext. There are also two main types of symmetric algorithms: Block ciphers and Stream ciphers:

- Block ciphers, such as Advanced Encryption Standard (AES), encrypt data in fixed-size blocks.
- Stream ciphers, encrypts one bit or byte at a time, making them reliable for a real-time data processing.

Asymmetric Key Encryption - Asymmetric Key Encryption or also known as public key cryptography basically means using two different keys—a private key and a public key—for encryption and decryption of the sensitive data. It is generally considered more secure but a little impractical than the Symmetric encryption method. This encryption works by making two keys one public which anyone can use to encrypt a plaintext and one private key which only the possessor of the respective private key of the public key can decrypt the ciphertext.

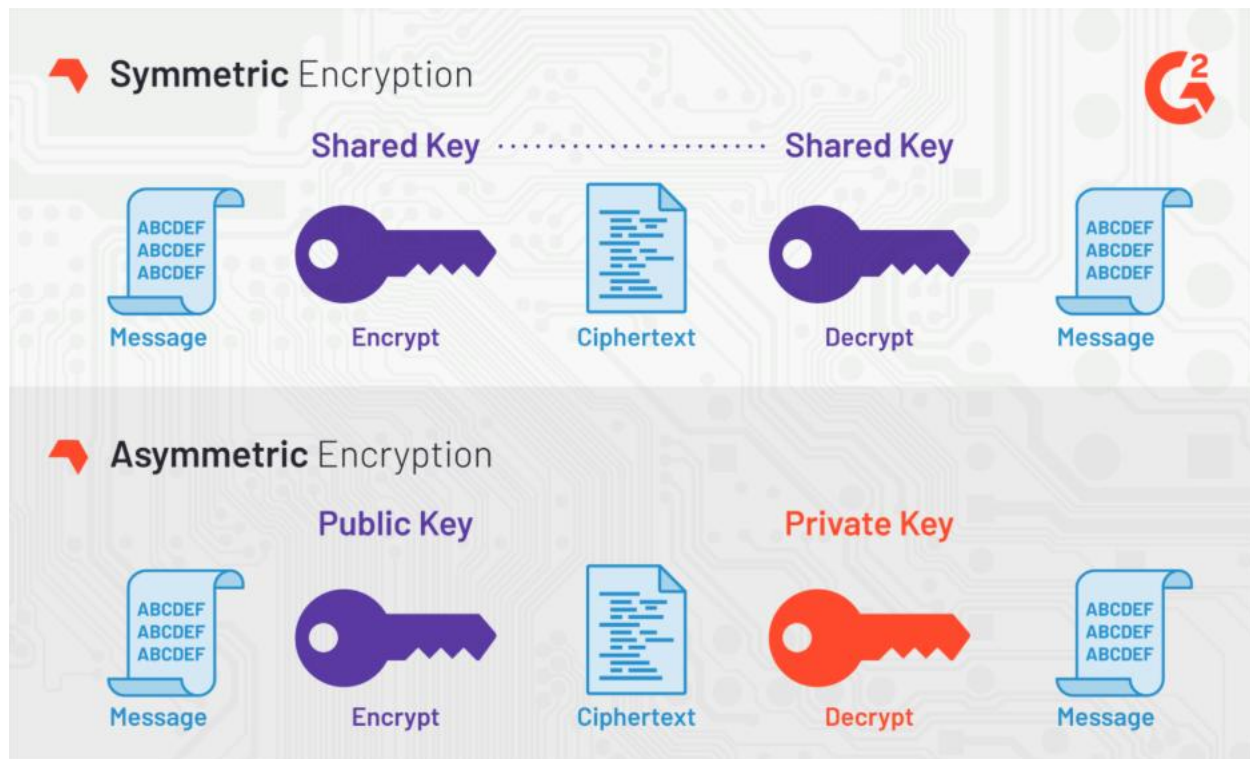


Figure 3: Symmetric and Asymmetric Encryption

1.3 History

Cryptography dates back thousands of years and was invented by the early civilizations to guard secret messages from enemies. The early civilizations applied very simple methods of guarding information in a manner that only the intended person was able to understand the meaning when decrypted. Scytale is one of the earliest cryptographic systems ever known, which was applied by the Spartans by writing messages on a paper which was wrapped around a wooden rod.

The classical era saw the emergence of cryptography in its organized form for secure communication. The most popular example of classical cryptography is the Caesar Cipher, which was used by Julius Caesar to protect his letters sent to his generals. The Caesar Cipher involved shifting each letter into the alphabet by a predetermined number of places. Although this method at that time proved to be effective, this form of cryptography is extremely simple and vulnerable to hacks.

During the medieval period, the art of cryptography also progressed along with the increasing complexity of international diplomacy and warfare. Substitution and transposition ciphers began being employed on a large scale for safeguarding confidential communication. It was during this time that the frequency analysis method of codebreaking was devised.

1.4 Aims and Objectives

1.4.1 Aims

The aim of this coursework is to understand the concept of cryptography and to study the history and evaluation of cryptography.

1.4.2 Objectives

- To study the basics of information system.
- To understand the concept of Confidentiality, Integrity and Availability.
- To research on the history of cryptography.
- To learn the difference between symmetric and asymmetric encryption.
- To study the working of Caesar cipher.
- To study the limitations, weaknesses and disadvantages of Cesar cipher.
- To research mathematical and logical operators for classical cipher.
- To design a new cryptography based on Caesar cipher.
- To apply both mathematical and logical operators on the new algorithm.
- To check the correctness of the new algorithm.
- To identify the weaknesses and limitations of the newly made cryptography.
- To understand the practical design of a cryptographic algorithm.

2. Background of Caesar Cipher

2.1 History

Caesar Cipher is one of the oldest encryption techniques in history. It was employed by Julius Caesar to transmit secret messages to his soldiers in case of any war. The basic intention of using the cipher technique was to ensure that if the messages were intercepted by the enemy, they would not be able to understand the messages.

This cipher operates by shifting alphabets by a predetermined number of places. For instance, by three places, the alphabet 'A' is coded as 'D'. Such simple cryptography was effective in the days when the world had little information regarding cryptography.

The Caesar Cipher was easily broken over time because it only has a few possible shifts to use. An attack on it was possible by trying the possible shifts available or by letter pattern deciphering. Due to this weakness, the Caesar Cipher is no longer used as an encryption technique.

Today, the Caesar Cipher is used as a teaching tool. The Caesar Cipher is used to teach students about the idea of encrypting and how secret keys are used. Although it is simple, the Caesar Cipher is significant because it laid the foundation for modern cryptography.

2.2 Working Mechanism

Caesar's Cipher works on the principle of shifting letters of a message to mask its meaning. In the method, each letter in the plaintext is shifted down or up fixed numbers of positions in the alphabet. This fixed number is called key or shift value.

First, a key is selected. The key determines the number of positions each letter will shift. Assuming the key selected is 3, the letter 'A' is replaced by 'D', 'B' goes to 'E', and 'C' turns to 'F'. Each letter of the message undergoes this shift.

On reaching the end of the alphabet, the shifting wraps around from the beginning. For instance, with a shift of 3: The letter 'X' becomes 'A'. The letter 'Y' becomes 'B'. The letter 'Z' becomes 'C'. This wrap-around behavior ensures that all letters will stay within the confines of the alphabet.

In the encryption process, the original readable message, known as the plaintext, is converted into an unreadable message called the ciphertext. Any person who does not know the value of the shift will not be able to easily understand the message. Decryption is the process that results in getting back the original message. In decryption, the same key is used, but this time, in the opposite direction, letters are shifted backward instead of forward. This reverse shifting brings them back to the original plaintext.

The Caesar Cipher comes down to a very simple rule of shifting, yet it is used for encryption and decryption of messages. Although it is really easy to understand and realize, it gives only the basic level of security and is now used exclusively for learning and educational purposes.

2.3 Advantages and Disadvantages

2.3.1 Advantages

One main benefit of the Caesar Cipher is that it is very easy to understand. The concept of shifting letters makes it simple for beginners to learn how encryption works. Because of its simplicity, it does not require complicated mathematics or advanced computing knowledge.

Another benefit is that the Caesar Cipher is easy to implement. It can be applied manually using pen and paper or implemented using a few lines of code. This makes it useful for educational purposes, especially for students who are new to the field of cryptography.

The cipher also helps in understanding basic cryptographic ideas such as plaintext, ciphertext, keys, encryption, and decryption. It provides a strong base for learning more advanced encryption techniques later.

2.3.2 Disadvantages

The biggest disadvantage of Caesar Cipher is that it is completely unsecured. The key range is very small, comprising a total of only 26 possible shifts. Due to this, the hacker may try all the keys effortlessly to break the cipher.

Another weakness is that the Caesar Cipher is based on very regular patterns: the letter frequencies in the plaintext are retained in the ciphertext, so frequency analysis attacks become very successful. This means that messages can be decrypted even without knowing the key.

The Caesar Cipher also cannot handle the demands for security that are made today. It provides insufficient security for sensitive data like passwords, finances, and personal records. As a result, it is no longer employed in practical security applications today but is used almost exclusively to learn from.

2.4 Weaknesses

- It is only 26 possible shifts (in the English alphabet), so it is possible to try all these options (brute-force attack) by an attacker.
- This cipher maintains the frequency of letters. Hence, the cipher is prone to frequency analysis attacks.
- It does not have the ability to protect personal data, passwords, financial information, and so on.
- In real-world implementations, it is hard to share and maintain the secret key.
- Even a beginner can easily decode it with no knowledge of the cipher key.
- Usually it works on letters only, without consideration of figures, symbols, or spaces.

3. Development of the New Cryptographic Algorithm

3.1 Research on Modification of Caesar Cipher

The traditional Caesar Cipher uses a fixed shift value applied uniformly to all characters of the plaintext. This uniformity, while making the cipher extremely simple, displaying it exceedingly vulnerable to cryptanalysis, such as, brute-force attacks or frequency analysis due to its predictable nature.

However, various modifications can be adapted to enhance security within the framework of the Caesar Cipher. These could include mathematical operations such as addition, multiplication, subtraction, and modulus operations that introduce additional complexity. Furthermore, logical operations based on conditions, like checking whether a character is in an odd or even position can be utilized to vary the character transformations. The integration of both mathematical and logical operations results in an encryption algorithm that is significantly more complex and secure compared to the traditional Caesar Cipher.

3.2 Background of the Mathematical and Logical Operations

In cryptography, mathematical operations are used in the process of transforming plain text into cipher text securely. In this algorithm addition and multiplication are used to calculate dynamic shift to determine the ASCII code values of character in the ciphertext. The cipher's final output is a number between 0 and the maximum possible ASCII code value, which is then reduced using the modulus operator.

The logical operations used in an encryption algorithm provide a means to introduce decision-making into the algorithm. In this algorithm Decision-making is based on the position of a character in the plaintext is accomplished using conditional statements, known as IF-ELSE statements. Depending on whether the character's position is odd or even, different dynamic shifts will be applied to that character. The use of logical decisions allows for multiple encrypted outputs from a single plaintext character.

3.3 Name of the New Cryptographic Algorithm

Name of the Algorithm:

LMS–Caesar Cipher (Logical–Mathematical Shift Caesar Cipher)

3.4 New Methodology Applied

The LMS–Caesar Cipher employs a position-centric method of encryption. That is, each character is coded based on its offset from a user-defined amount, which is derived from both its position within the plaintext and the user's selection of a shift amount (i.e. key). Logical relationships are used in combination with mathematical procedures to establish the shift for each character and alter the ASCII value associated with each character. A reversing operation completes the process and provides additional obscurity of the encoded text.

3.5 New Encryption Algorithm

The steps for encrypting a message using the LMS Caesar Cipher are as follows:

- Take the plaintext message and a numeric key K .
- Convert each character of the plaintext into its respective ASCII value.
- Assign a position number P to each character (starting from 1).
- Apply a logical condition:
 - If $P(\text{Position})$ is odd, calculate $\text{shift} = K + P$
 - If P is even, calculate $\text{shift} = (K \times 2) - P$
- Add the calculated shift to the ASCII value of the character.
- Apply mod 256 to keep the ASCII value within range.
- Convert the new ASCII value back to a character.
- After all characters are encrypted, reverse the final ciphertext.

3.6 New Decryption Algorithm

The steps for decrypting a message using the LMS Caesar Cipher are as follows:

- Take the ciphertext and the same numeric key K .
- Reverse the ciphertext back to its original order.
- Convert each character into its ASCII value.
- Assign position numbers P starting from 1.
- Apply the same logical condition used in encryption:
 - If P is odd, subtract $(K + P)$

- If P is even, subtract $(K \times 2) - P$
- Apply mod 256 to obtain the original ASCII value.
- Convert the ASCII value back to its original character.
- Combine all characters to retrieve the original plaintext.

“The pseudocodes for both Encryption and Decryption of LMS-Ceaser Cipher is given in the Appendices section.”

3.8 Flowchart

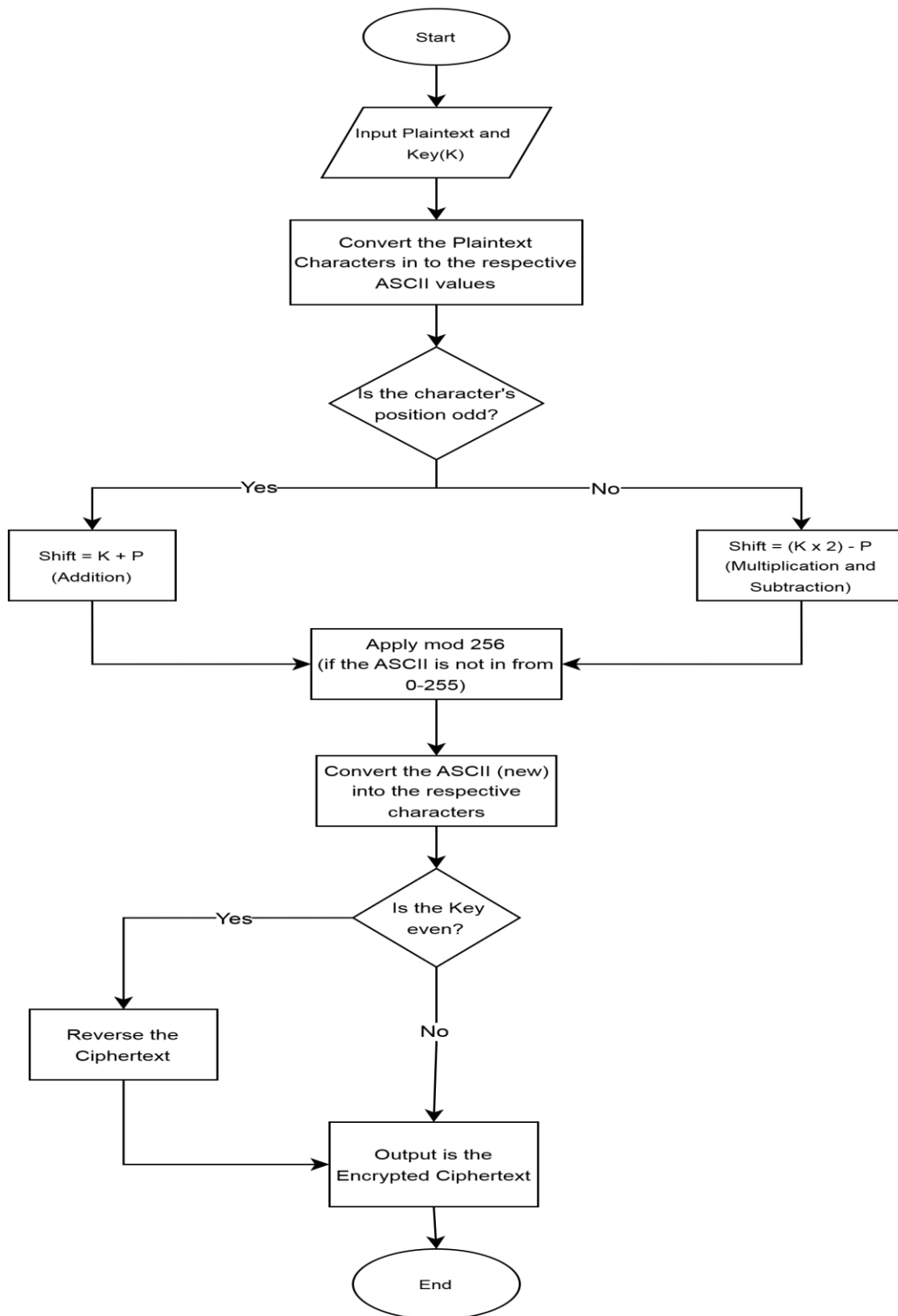


Figure 4: Encryption Process Flowchart

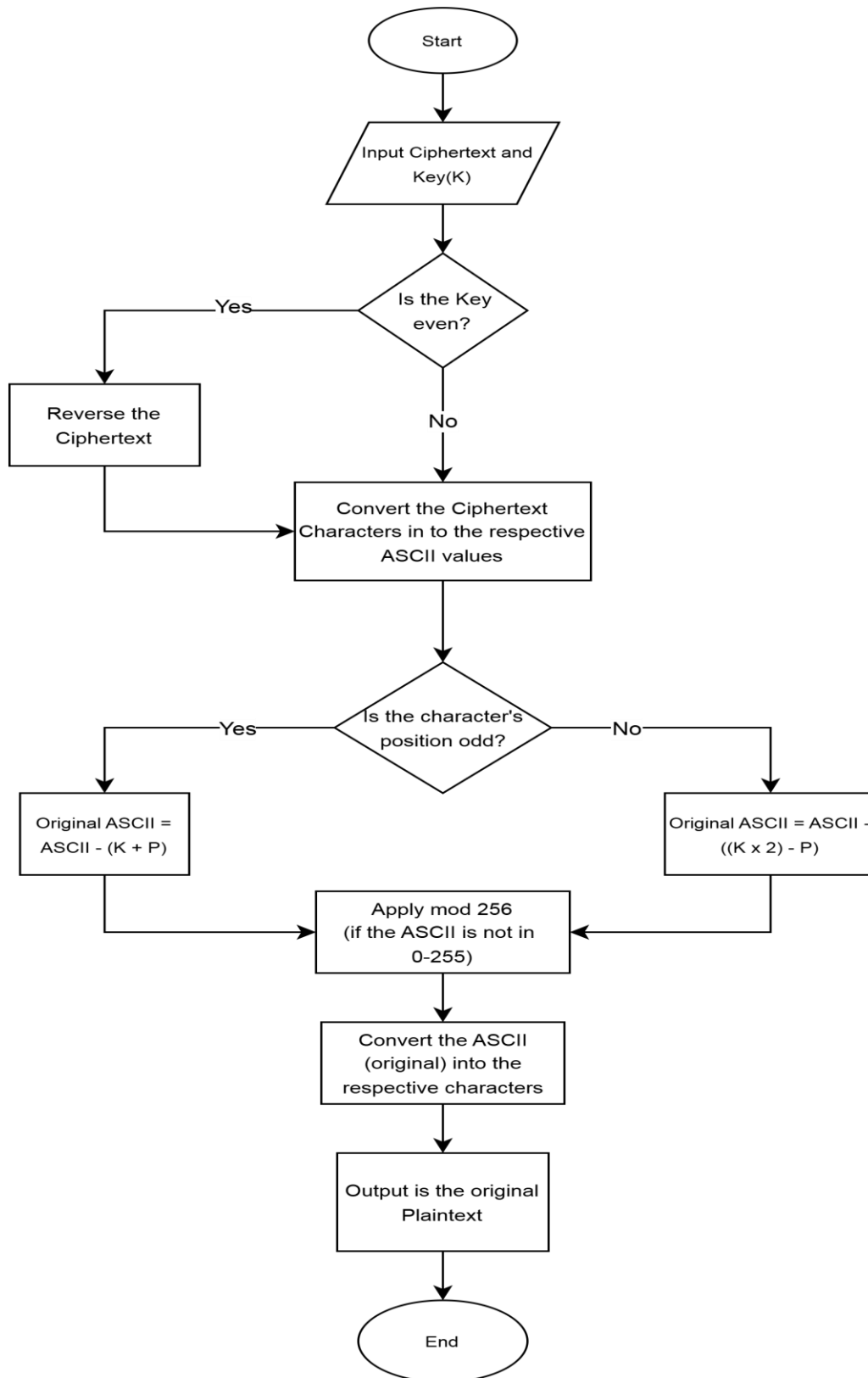


Figure 5: Decryption Process Flowchart

The flowchart of LMS-Caesar Cipher starts with receiving a plaintext and a key as input. After converting the plaintext to ASCII Code and checking its position, logical conditions must be satisfied to determine how much a character should be shifted. Mathematics are then used to create each encrypted character. Finally, the encrypted string of characters is reversed and presented as output. When creating an encrypted version of the original plaintext, the same flowchart can be used, but in reverse.

4. Testing

4.1 Overview

In this section we can see how the LMS–Caesar Cipher works through a number of examples that show you how to test it. Each of the five different plaintext inputs will be encrypted using the LMS–Caesar Cipher, then decrypted back to ensure we are able to recover the original plaintext. By completing these test cases, we are verifying that the algorithm can provide an accurate and dependable means of encrypting and decrypting messages.

Test Case 1

Plaintext: HELLO

Key: 3

Test OBJECTIVE: To test the LMS Ceaser Cipher.

Encrypted Text: LIRNW

Test Conclusion: The test was successful.

Test Case 2

Plaintext: WORLD

Key: 4

Test OBJECTIVE: To test the LMS Ceaser Cipher.

Encrypted Text: QZSMF

Test Conclusion: The test was successful.

Test Case 3

Plaintext: CRYPTO

Key: 5

Test OBJECTIVE: To test the LMS Ceaser Cipher.

Encrypted Text: HWDZXT

Test Conclusion: The test was successful.

Test Case 4

Plaintext: SECURE

Key: 6

Test OBJECTIVE: To test the LMS Ceaser Cipher.

Encrypted Text: VJFWXH

Test Conclusion: The test was successful.

Test Case 5

Plaintext: TESTING

Key: 7

Test OBJECTIVE: To test the LMS Ceaser Cipher.

Encrypted Text: ZLZQVO

Test Conclusion: The test was successful.

5. Analysis of the LMS-Ceaser Cipher

5.1 Overview

This section evaluates the strengths & limitations of a newly developed encryption method, called the LMS-Caesars Cipher that was developed in Task 03. This evaluation shows how the logical & mathematical improvements made to this method increase its security beyond the traditional Caesar Cipher but also addresses any limitations that these improvements have.

5.2 Strengths of the LMS-Ceaser Cipher

- By combining logical and mathematical processes, this method enhances overall security compared to the original Caesar Cipher method of encryption.
- Instead of having a fixed shift value, this method changes the shift based on where each character falls within the plaintext message, thereby increasing the unpredictability of shifts during the encrypting process.
- It is possible for one character to produce many unique encrypted outputs based on where it is in the plaintext message. For example, if an “A” appears at the beginning of a plaintext message, the output will differ from an “A” that appears at the end of the message.
- Using ASCII Values provides many different types of letters, numbers, and special characters to manipulate in this encryption method.
- This encryption method is very easy to understand and can be applied to educational and academic studies as well.

5.3 Weaknesses of the LMS-Ceaser Cipher

- While this encryption algorithm is stronger than the classic Caesar cipher, it does not provide a sufficient level of strength for use in today's highly secure applications.
- This algorithm's reliance upon one numeric key result in limited potential complexity of that numeric key.
- If an analyst discovers the algorithm's logic, that analyst can reverse the encrypted text with further deep analysis.
Encryption tends to slow down for massive text data input, relative to the less complex ciphers.

- Additionally, the Coupling Cipher algorithm does not implement more sophisticated forms of encryption technology such as hashing or key exchange mechanism.

Conclusion

The goal of this project was to research, design, implement, and evaluate a modified version of the original Caesar Cipher called the LMS-Caesar Cipher. This project begins by examining the original Caesar Cipher for its limitations - primarily on how it functions using a single fixed shift value and is vulnerable to frequency analysis. In order to overcome those limitations, the LMS-Caesar Cipher utilizes mathematical and logical operations to provide a level of complexity to a user's key.

The LMS-Caesar Cipher demonstrated that adding a few simple techniques can greatly increase the overall strength and unpredictability of an encryption method, while still maintaining a lightweight design and low implementation costs. The author provided flowcharts and pseudocode to support the LMS-Caesar Cipher's encryption and decryption workflows. This has provided clear documentation that will allow examiners to verify that the encryption and decryption processes are working properly and are reproducible by all testers. Additionally, numerous test case analyses confirmed that the LMS-Caesar Cipher could consistently revert an encrypted message back to its original plaintext; thus, verifying the strength of the algorithm.

The LMS-Caesar Cipher satisfied its goals of improving the security of a standard classical cryptosystem and demonstrated the ability to utilize both logical and mathematical operators to achieve this goal. Although there are many encryption algorithms considered to be more secure than the LMS-Caesar Cipher, the algorithm demonstrates an important part of cryptographic design and how reliable and continuous improvements must be made to existing encryption technologies.

Appendices

Appendices 1: Differences between Symmetric and Asymmetric Encryption

Symmetric encryption is different from asymmetric encryption, because it uses only one key for both encryption and decryption of data, whereas asymmetric encryption uses two keys, one public and one private to encrypt and decrypt the data.

Using the same key for encryption and decryption in symmetric key means it is faster and easier to use but it can also be vulnerable to many threat actors. Symmetric encryption requires two parties to share the key which in between exchanging can also be accessed by a hacker. Looking at this case it is more secure to use asymmetrical encryption methods.

Appendices 2: Reason for Modification of Ceaser Cipher

This modification to the original Caesar Cipher algorithm was created due to the very low level of security associated with it. The Caesar Cipher's algorithm consists of a simple, fixed shift with no consideration given to any logical conditions and/or advanced mathematical operations. Thus, with the incorporation of methods from both logical and mathematical techniques in the new algorithm, it has become much less predictable than before, making the algorithm less susceptible to the simplest types of cryptographic attacks.

Appendices 3: Pseudocode for Encryption of LMS-Caeser Cipher

START

INPUT plaintext

INPUT key K

SET position P = 1

SET ciphertext = empty

FOR each character C in plaintext DO

 CONVERT C to ASCII value A

 IF P is odd THEN

$\text{shift} = K + P$

 ELSE

$\text{shift} = (K * 2) - P$

 ENDIF

$\text{newValue} = (A + \text{shift}) \text{ MOD } 256$

 CONVERT newValue to character E

 ADD E to ciphertext

 INCREMENT P by 1

END FOR

REVERSE ciphertext

OUTPUT ciphertext

END

Appendices 3: Pseudocode for Decryption of LMS-Caeser Cipher

START

INPUT ciphertext

INPUT key K

REVERSE ciphertext

SET position P = 1

SET plaintext = empty

FOR each character C in ciphertext DO

 CONVERT C to ASCII value A

 IF P is odd THEN

$\text{shift} = K + P$

 ELSE

$\text{shift} = (K * 2) - P$

 ENDIF

$\text{originalValue} = (A - \text{shift}) \text{ MOD } 256$

 CONVERT originalValue to character D

 ADD D to plaintext

 INCREMENT P by 1

END FOR

OUTPUT plaintext

END

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